Abdullah Khan

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Education

University of Waterloo & Wilfrid Laurier University — Waterloo, ON
Double Degree: Bachelor of Computer Science (BCS) & Bachelor of Business Administration (BBA) — *Expected 2029*
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Linear Algebra, Statistics, Probability

Experience

Software Engineering Intern — Aurora, ON
LeapAP Inc. — *May 2026 – Aug. 2026*

- Architected a config-driven **Node.js/Puppeteer** scraper framework that reduced each new billing integration to 3 config changes, cutting build time from **3+ hours to under 10 minutes**; adopted as the base for 11 production scrapers.
- Engineered **32+ scrapers and 28+ parsers** for utility, telecom, and waste provider portals, expanding automated invoice coverage across 3 industries and closing 185+ tickets in 4 months with daily production deploys via GitLab.
- Rebuilt the scraper for **a major Canadian telecom provider**, now pulling invoice data for **400+ customer accounts**.
- Restored data collection across **100+ billing portals**, including scrapers serving 2,700+ and 2,200+ accounts, by diagnosing and fixing **110+ production failures** caused by selector changes, MFA flows, and CAPTCHAs using Grafana and Graylog.
- Owned a **biweekly audit** of false "account not found" and "bad credentials" errors, tracing 10+ misclassified failures to their root cause and correcting them so error reports stayed reliable across every scraper and parser integration.

Software Developer — Milton, ON
PixelsBoost — *Sep. 2025 – Dec. 2025*

- Engineered **5 full-stack client websites** from designer mockups in React, JavaScript, and HTML/CSS, with dynamic contact forms, interactive image galleries, and mobile-responsive navigation for small-business clients.
- Integrated **3 third-party APIs** (Stripe payments, Google Maps geolocation, SendGrid email automation), adding error handling and retry logic to keep checkout and contact flows reliable when external services failed or timed out.
- Raised Google Lighthouse performance scores from **68 to 87** by implementing WebP image compression, lazy loading, and Cloudflare CDN caching, speeding up page loads on image-heavy client sites for both mobile and desktop visitors.

Software Engineering Intern — Missouri, USA (Remote)
Fast Webs — *May 2024 – Aug. 2024*

- Developed **12 production React components** for 3 Node.js web applications, including checkout flows, user dashboards, and admin panels built on a reusable component architecture shared across all 3 client-facing applications.
- Built an **event-tracking system** capturing clicks, form submissions, and navigation events into PostgreSQL, giving the product team behavioral data that surfaced 3 UX improvements for the team to prioritize in upcoming sprints.
- Reduced JavaScript bundle size by **120KB** through React code splitting and lazy loading, profiled with webpack bundle analyzer to find the heaviest imports, shortening initial page loads across all 3 production web applications.

Projects

Bill Extractor | GitHub | *Python, Claude API, Pydantic, PostgreSQL, pgvector, Docker, pytest*

- Built an **LLM extraction pipeline** (Claude API, Pydantic, PyMuPDF) that turns utility bill PDFs into schema-validated data, extracting **1,000/1,000 fields** correctly on a held-out set of 100 synthetic bills at $0.0028 and 2.45s per bill.
- Proved LLM extraction generalizes where templates fail: hand-written regex parsers hit 100% on their 2 target layouts but **under 9%** on 3 unseen ones; added arithmetic checks with model-feedback retries and 236 pytest tests.
- Added **RAG** Q&A with pgvector (HNSW) retrieval and text-to-SQL routing (30/30 routing, 13/13 recall@5), securing generated SQL with sqlglot AST validation, a SELECT-only PostgreSQL role, and scoped read-only transactions.

BookPulse | Live | GitHub | *React, TypeScript, FastAPI, PostgreSQL, Docker, AWS (Lambda, S3, RDS, ECS, EventBridge)*

- Built a full-stack **book price intelligence platform** tracking 100+ books, with a React/TypeScript frontend on Vercel and a Dockerized FastAPI backend on AWS ECS Fargate backed by RDS PostgreSQL, secured with IAM roles and CORS.
- Architected a **serverless ingestion pipeline** where EventBridge triggers AWS Lambda hourly to store raw data in S3 and compute price-change, demand, and rarity scores across 5 categories for a live Chart.js dashboard.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, Go, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, Spring Boot, Tailwind CSS, Puppeteer, Pandas, NumPy
AI & LLMs: OpenAI & Anthropic APIs, RAG, LangChain, LangGraph, AWS Bedrock, MCP, Agentic Workflows, Prompt Engineering
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Tools & Cloud: AWS (Lambda, S3, RDS, ECS), Docker, Git, GitLab, Jira, Linux, Grafana, Graylog, Vercel, pytest, Power BI, Tableau